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92/100

Q1) a) Define Data Mining

Data mining is the Process of extracting useful Patterns, knowledge, and information from large amount of data

Any Two Data Mining Tasks

* classification - classifies data into Predefined classes
* clustering - Groups similar data without Predefined labels

steps in KDD Process

* Data cleaning
* Data Integration
* Data selection
* Data Transformation
* Data mining

Role of Data processing

* Removes errors and missing values
* improves data quality
* reduces processing time

b) Data cleaning

Data cleaning is the Process of removing incorrect, incomplete or inconsistent data.

Two data quality issues

* Missing values
* Noisy data

Handling missing values.

* Replace with mean/median.
* fill manually
* ignore records

Ex Age = ?, replace with average age

salary = ₹ 10,00,000 among ₹ 20,000

(2) a) Data Integration

Data integration combines data from multiple sources into one dataset

challenges

1. Data redundancy
2. Data inconsistency

Handling inconsistency

* standardize formats
* correct conflicting values

b) covariance Analysis

covariance measures how two variables change together

Two covariance forms

1. Poistive covariance
2. Negative covariance

(3) a) given

A = (2,3,5,4,6)

B = (5,8,10,11,14)

As stock A increases, stock B also increases.

a) Equal - Frequency Binning

Data.

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 33,
35, 35, 35, 36, 40, 45, 46, 52, 70

Total = 25 values

Bin 1 : 13, 15, 16, 16, 19

Bin 2 : 20, 20, 21, 22, 22

Bin 3 : 25, 25, 25, 30, 33

Bin 4 : 33, 35, 35, 35, 36

Bin 5 : 40, 45, 46, 52, 70

b) Data Discretization

Data Discretization converts continuous values into intervals

Equal - width Binning

* Same interval width

* Example:

0 - 20

21 - 40

41 - 60

Equal width	Equal frequency
Same interval size	Same no of values
Simple	Handles skewed data better.
May have uneven data	Balanced bins

b) Architector of Data Mining system

data sources
↓
Database / Data warehouse
↓
Data cleaning & Integration
↓
Data Mining Engine
↓
user interface

<u>Functions</u>

Data Sources → store raw data

Database → organizes data

preprocessing → cleans data

4) a) given

$x = \{4, 5, 6, 9\}$ $y = \{8, 3, 7, 5, 6\}$

mean of $x = 6.2$ mean of $y = 5.8$

Covariance

$= -0.85$

b) major steps in pre processing

* Data cleaning
* Data Integration
* Data Transformation
* Data Reduction

Pipeline

Missing values Remove Duplicates ↓ Handle outliers ↓ standardize formats ↓ Transform Data ↓ final Dataset.